

HW 1

Problem 1

Let G be a group and $N \trianglelefteq G$

Let ψ be an homomorphism from G to G_1 s.t. $N \subseteq \ker(\psi)$

Define: $\bar{\psi} : G/N \rightarrow G_1$ s.t. $\bar{\psi}(gN) = \psi(g)$

Well Defined: Suppose $gN = hN \implies gh^{-1} \in N \implies gh^{-1} \in \ker(\psi)$

$\psi(gh^{-1}) = e \implies \psi(g) = \psi(h)$ since ψ is a homomorphism this implies $\bar{\psi}(gN) = \bar{\psi}(hN)$

Homomorphism: Let $gN, hN \in G/N$

$\bar{\psi}(gNhN) = \bar{\psi}(ghN) = \psi(gh) = \psi(g)\psi(h) = \bar{\psi}(gN)\bar{\psi}(hN)$

Uniqueness: Suppose there exists another homomorphism $\phi : G/N \rightarrow G_1$ such that $\phi = \psi \circ \varphi$

$$\phi(gN) = \psi(g) \implies \phi = \bar{\psi}$$

Problem 2

Let H be a subgroup of G .

- (Inverses) Let $a \in N_G(H)$
Consider: $a^{-1}Ha = a^{-1}aH = H$ thus $a^{-1} \in N_G(H)$
- (Closure) Let $a, b \in N_G(H)$
Consider: $abHb^{-1}a^{-1} = abb^{-1}Ha^{-1} = aHa^{-1} = H$ thus $ab \in N_G(H)$
- (Identity) Consider $eHe = H$ thus $e \in N_G(H)$

Let $h \in H$

Consider : hHh^{-1} since $h, h^{-1} \in H \implies hHh^{-1} = H$ thus $h \in N_G(H)$ thus $H \subseteq N_G(H)$

By definition since $\forall g \in N_G(H) gHg^{-1} = H$ thus $H \trianglelefteq N_G(H)$

Largest: Let K be a subgroup of G s.t. $H \trianglelefteq K$ thus $\forall k \in K$

$$kHk^{-1} = H$$

thus $k \in N_G(H)$ thus $K \subseteq N_G(H)$

Problem 3

- (i. $\implies$ ii)

Proof. Assume i.

Let $g \in H \cap K \implies g \in H$ and $g \in K$

□